Biochemistry Cume April 29, 2016

Nucleic acids

1. (15 pts) If a double-stranded DNA genome contains 33% G, what is its complete base composition?

2. (10 pts) If a single-stranded DNA genome contains 33% G, what is its complete base composition?

3. (8 pts) Which of the following could change the topological linking number L of a 13,500-base pair covalently closed supercoiled DNA molecule (indicate all that apply):

   A) DNA gyrase with ATP?
   B) heating to 50 °C?
   C) heating to 95 °C?
   D) heating to 95 °C, then cooling to 45 °C?
   E) breaking and then rejoining one phosphodiester bond on one strand?
   F) breaking and then rejoining one phosphodiester bond on each of the two strands?
   G) binding a lysine-rich protein?
   H) binding a primer of single-stranded RNA?

4. (8 pts) Explain all of your answers to #3 (both the "no" and the "yes" answers).

5. (5 pts) The energy for the elongation of RNA by polymerase comes primarily from _____.

6. (10 pts) Explain your answer to #5. Be specific. Explain why this energy source also renders the elongation reaction irreversible.

7. (8 pts) T and not U is found in DNA because (indicate all that apply):

   A) G will not pair with U.
   B) A will not pair with U.
   C) C converts slowly to U.
   D) repair systems remove U from DNA.
   E) deoxyribose is unstable with U.
   F) C deaminates.
   G) A may become methylated.
   H) transcription could not occur without T.

8. (16 pts) Explain each of your answers to #7 (both the "no" and the "yes" answers).

9. (5 pts) Which of the following is more easily hydrolyzed by alkali? (Indicate the <u>best</u> answer.)

   A) RNA.
   B) DNA
   C) Both RNA and DNA.
   D) Neither RNA nor DNA.

10. (15 pts) Explain your answer to #9. Include a chemical mechanism that shows as much detail as time and knowledge permit.